EXPERIMENT 9 (A)

18/9/24 IMPLEMENTATION AND SUBNETTING OF CISCO PACKET TRACER

Aim

To design and implement subnets using Cisco Packet tracer, assign static IP address to host connecting subnets, using a router simulator, simulate Packet transmission.

Algorithm:

Step ①: Open Cisco Packet tracer

Step ②: Create subnets LAN / subnet using PC, switch.

Step ③: Select a suitable network address and divide into subnet.

Step ④: Assign static IP address, subnet mask, default gateways

Step ⑤: Connect each switch to the interface

Step ⑥: Configure the router interface with appropriate IP address.

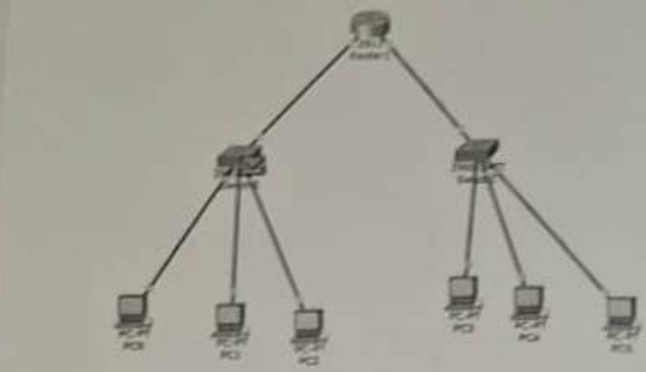
Step ⑦: Verify using PING

Step ⑧: Use simulation mode to send a Packet from PC to another

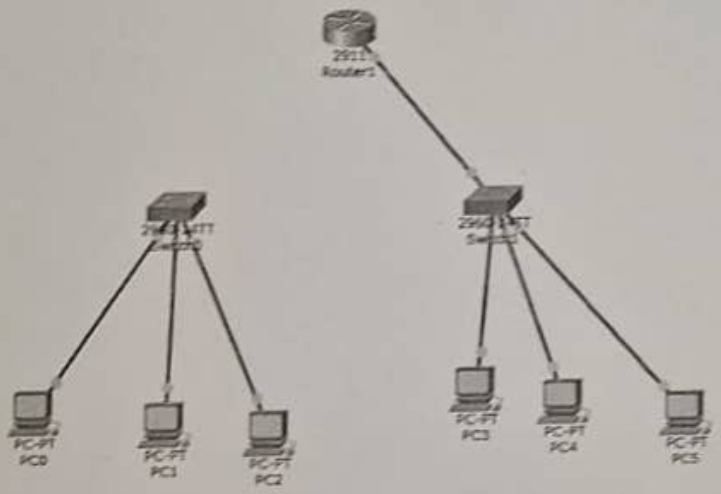
Step ⑨: Observe Packet transmission through router

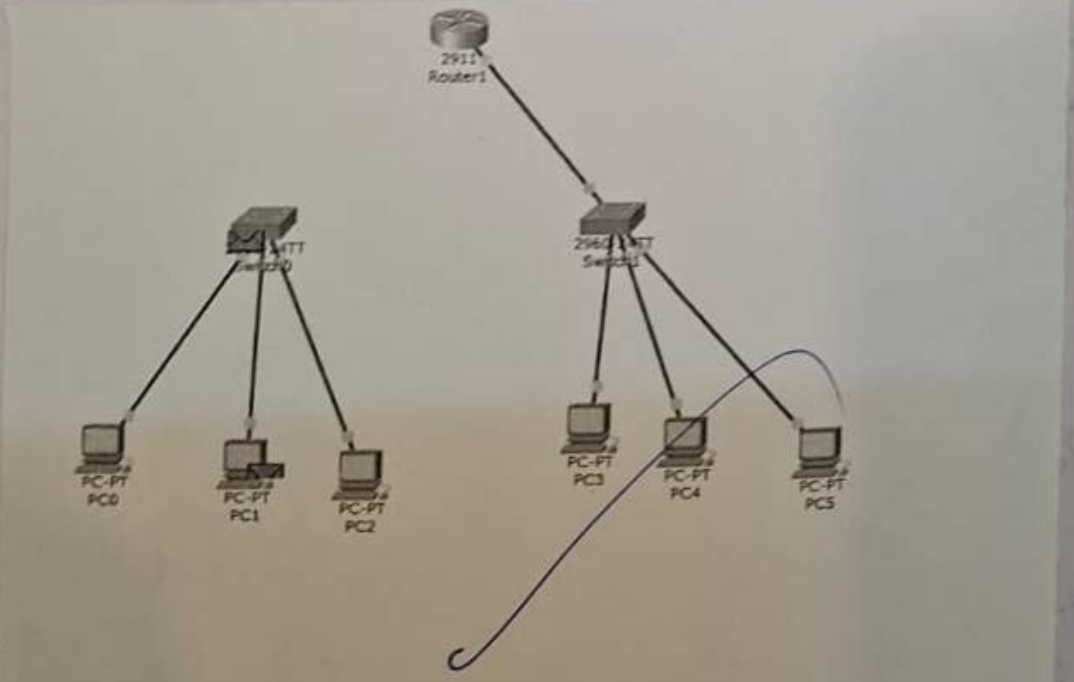
Step ⑩: Remove / disconnect router, connect

Step ⑪: Compare result when subnet are connected



Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)

Procedure :

① Create network topology

* 1 Router - 2911
* 4 PC
* 2 switch - 2960

② Create 2 subnet

Subnet ①

network address : 192.168.1.0

subnet mask : 255.255.255.192

usable range : 192.168.1.1 - 192.168.1.62

Broadcast : 192.168.1.63

Subnet ②

network address : 192.168.1.64

subnet mask : 255.255.255.192

usable range : 192.168.1.65 - 192.168.1.126

Broadcast. : 192.168.1.127.

③ Assign IP address to PC

④ Configure router - CLI

Enable

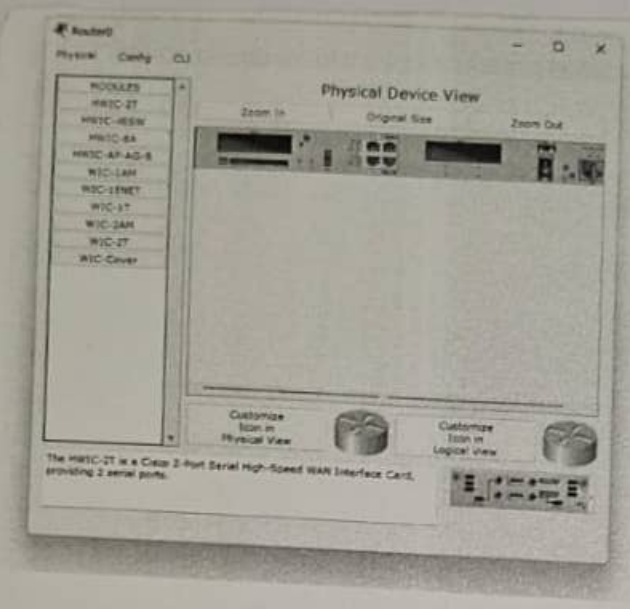
Configure terminal

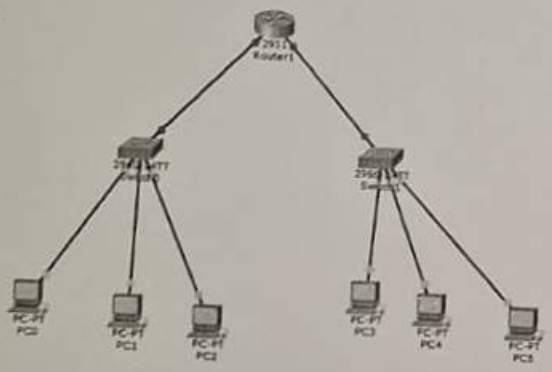
interface gigabit, Ethernet 0/0.

IP address 192.168.1.1 255.255.255.192

no shutdown

exit






```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```

Copy Paste

⑤ Test communication within same subnet

Run : Ping 192.168.1.11

Expected : reply from 192.168.1.11

reply from 192.168.1.11

∴ PC0 and PC1 can communicate within same subnet

⑥ Communicate between different subnet

Run : Ping 192.168.1.70

PC0 : 192.168.10.10/24

PC2 : 192.168.1.64/24

⑦ Simulate Packet Transmission.

* add simple PDU

PC0 ⟶ PC2

PC0 ⟶ Switch → Router ⟶ Switch 1 → PC2

⑧ Test without a route

Now, disconnect the router from switch

Try PC0 ⟶ PC2

(or) from PC0 → Ping 192.168.1.70

The communication should fail, since the routers are disconnected from switch.

Result:

Thus, the network was manually subnetted and configured in Cisco Packet Tracer using 2 subnets, static IP address are assigned to all PC's and subnets interconnected using a router.

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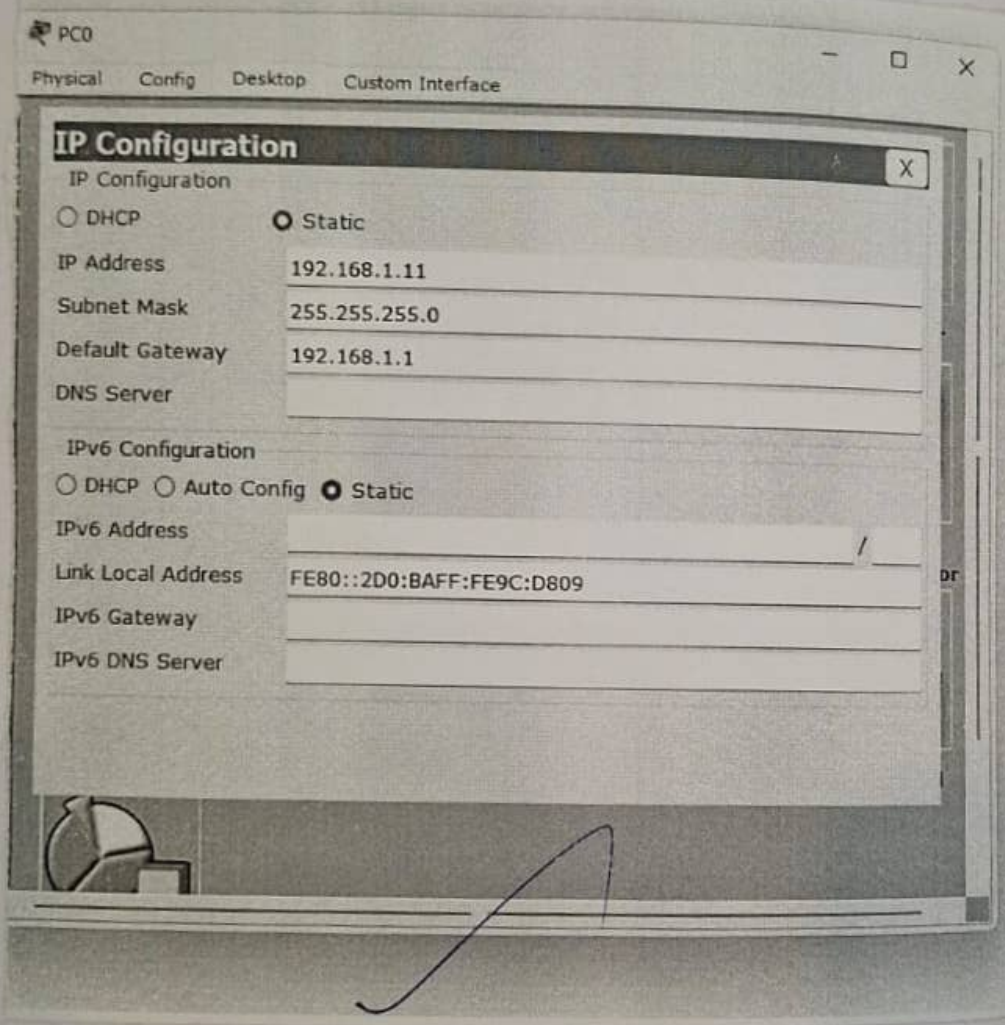
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PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
IP Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::2D0:BAFF:FE9C:D809
IPv6 Gateway
IPv6 DNS Server

a (b)

12/6/26 ASSIGN STATIC IP ADDRESS ACROSS ALL SUBNET AND CONNECTS THE SUBNET THROUGH A ROUTER

AIM: To assign static IP address to all devices in device in different subnet and establish communication between them the subnet through a router.

Step:

(i) Plan a router in network topology
(ii) Connect the router to the switch using appropriate cables
(iii) Configure the router interface
(iv) Assign an IP address and subnet mask to each other router interface
(v) Configure the switches
(vi) Assign static IP address to all computers interface in every subnet
(vii) Configure the subnet mask for each device
(viii) Assign the appropriate default ip gateway to each computer.
(ix) Verify the IP address configuration of all device
(x) Save the router configuration
(xi) Check the network topology for configuration errors
(xii) Verify that all device are properly connected.

Procedure:

① Plan the wiring
- Open Cisco Packet Tracer
- Place on router, the required switches and computers (PC's) for each subnet

② Connect the wiring
- Connect each PC to its respective switch using copper straight through cables
- Connect each switch to a different router interface using copper straight-cables

③ Configure the router
- Open the router and using copper-straight through cables

④ - Enter privileged mode enable
- Configure terminal of configuration mode

Example
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit.

⑤ Configure the switches
Connect the PC's to the appropriate switch port
Ensure the switch port are active and working correctly.

⑤ Assign static IP address to PCs

- Open each PC
- Go to desktop → IP configuration
- Select static

Enter the appropriate

IP address
Subnet mask
default gateway

⑥ Configure the switches

- open each PC
- Go to desktop - IP configuration
- select static

Enter the appropriate

(i) IP address
(ii) Subnet mask
(iii) default gateway

⑦ Verify the configuration

the IPconfiguration check that IP addr and default gateway are correct.

⑧ Test connectivity

from each PC then ping it default gateway ping 192.168.1.1 and Ping 192.168.2.2

⑨ Check the router

In the router CLI use

Show ip interface brief

Verify that the configured interface

9. Check the router

- In router cli, use:
  Show ip interface brief; verify all up/up and and have the connection IP address

10. Save the router cli use;
  Show ip interface brief
  Verify that configured interfaces are up/up and have correct IP address

11. Check PC for errors
- Verify all cables and interface connection
- Confirm the subnet mask and default gateway correct.
- Correct any configuration

12. Final verification
- Ping between PCs in different subnet
- If the pings success, the network is correctly configured.

Result:

Thus, static IP address were successfully assigned to all devices and communication between subnet was established through the router router.